

241001235 - Sathyaseelan.C

8 x 8 Queen's Problem

Program:

```
N=8
board = [[0 for i in range(N)] for i in range(N)]

def is_safe(board,row,col):

    for i in range(col):
        if board[row][i]=='Q':
            return 0

    i=row
    j=col
    while i>=0 and j>=0:
        if board[i][j]=='Q':
            return 0
        i-=1
        j-=1

    i=row
    j=col
    while i<N and j>=0:
        if board[i][j]=='Q':
            return 0
        i+=1
        j-=1

    return 1

def solve(board,col):
    if col>=N:
        return 1
```

```

for row in range(N):
    if is_safe(board,row,col):
        board[row][col]='Q'

        if solve(board,col+1):
            return 1

        board[row][col]=0

return 0

if solve(board,0):
    for row in board:
        print(row)
else:
    print("no solution")

```

Output:

```

Output
['Q', 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 'Q', 0]
[0, 0, 0, 0, 'Q', 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 'Q']
[0, 'Q', 0, 0, 0, 0, 0, 0]
[0, 0, 0, 'Q', 0, 0, 0, 0]
[0, 0, 0, 0, 0, 'Q', 0, 0]
[0, 0, 'Q', 0, 0, 0, 0, 0]

```